HEALTHCONNECT RWANDA

Week 6 Healthcare Operations & Performance Analytics

AnalystLab Africa Experience Lab | Data Analytics Track

Executive Summary

This package establishes the analysis framework for the Week 6 HealthConnect Rwanda assignment. The official brief specifies 27 interconnected healthcare datasets and requests SQL analysis, a six-page Power BI executive dashboard, a business insights report and an executive presentation. The healthcare data files or database backup were not available in the files accessible during preparation, so numeric findings and facility rankings cannot yet be calculated responsibly.

Scope from the official brief

The work covers patient demographics and insurance, appointment demand, clinical activity, staffing, facility operations, billing and claims, revenue, and patient experience. The required dashboard pages are Executive, Patient, Clinical, Financial, Operational, and Patient Experience.

Analysis plan

1. Inventory all received tables and validate keys, missingness, duplicates, date fields, categorical values and relationship integrity.
2. Use PostgreSQL to answer the 40 patient, appointment, clinical and financial questions in the accompanying SQL script.
3. Build KPI definitions with explicit denominators, time windows and business rules.
4. Reconcile financial totals across billing, payments, claims and revenue summary.
5. Build the six Power BI pages, test filters and drill-through, and reconcile every KPI against SQL outputs.
6. Translate validated patterns into operational recommendations and document limitations.

Data and ethical safeguards

Healthcare data is sensitive. Use only authorized, de-identified records; restrict access; do not publish patient names, phone numbers, addresses, or identifiers. Avoid clinical claims about disease trends or patient outcomes unless the data definitions, dates, coding and denominators are verified. Do not interpret billed revenue as collected or recognized revenue without finance-approved definitions.

Deliverable status

Deliverable	Prepared status
PostgreSQL SQL script	Starter queries for all 40 listed questions
KPI workbook	Dictionary and validation checklist
Power BI build guide / DAX	Prepared; adapt to actual schema
Executive report	Framework prepared; numeric findings pending data
Presentation	Framework prepared; numeric findings pending data
Native .pbix and database backup	Not created: requires actual data and local Power BI/PostgreSQL environment

Next step required for final evidence-based results

Add the actual 27 healthcare datasets or PostgreSQL database backup. Once available, the SQL templates can be mapped to the real schema, executed, and used to populate the final report and dashboard with verified values.